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MATHEMATICS

0580/22

Paper 2 Non-calculator (Extended)

May/June 2025

2 hours

You must answer on the question paper.

You will need: Geometrical instruments

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- Calculators must **not** be used in this paper.
- You may use tracing paper.
- You must show all necessary working clearly.

INFORMATION

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [].

This document has **20** pages. Any blank pages are indicated.

List of formulas

Area, A , of triangle, base b , height h .

$$A = \frac{1}{2}bh$$

Area, A , of circle of radius r .

$$A = \pi r^2$$

Circumference, C , of circle of radius r .

$$C = 2\pi r$$

Curved surface area, A , of cylinder of radius r , height h .

$$A = 2\pi rh$$

Curved surface area, A , of cone of radius r , sloping edge l .

$$A = \pi rl$$

Surface area, A , of sphere of radius r .

$$A = 4\pi r^2$$

Volume, V , of prism, cross-sectional area A , length l .

$$V = Al$$

Volume, V , of pyramid, base area A , height h .

$$V = \frac{1}{3}Ah$$

Volume, V , of cylinder of radius r , height h .

$$V = \pi r^2 h$$

Volume, V , of cone of radius r , height h .

$$V = \frac{1}{3}\pi r^2 h$$

Volume, V , of sphere of radius r .

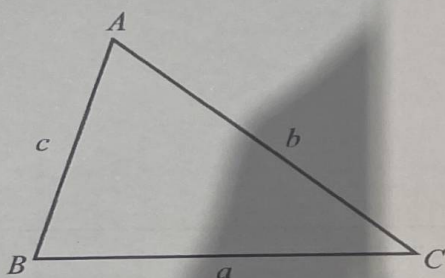
$$V = \frac{4}{3}\pi r^3$$

For the equation

$$ax^2 + bx + c = 0, \text{ where } a \neq 0,$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For the triangle shown,



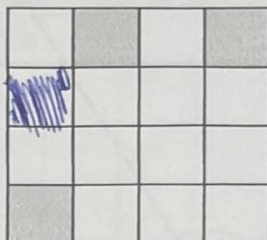
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{Area} = \frac{1}{2}ab \sin C$$

Calculators must **not** be used in this paper.

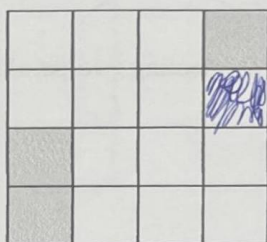
1 (a)



Shade **one** more small square so that the diagram has one line of symmetry.

[1]

(b)

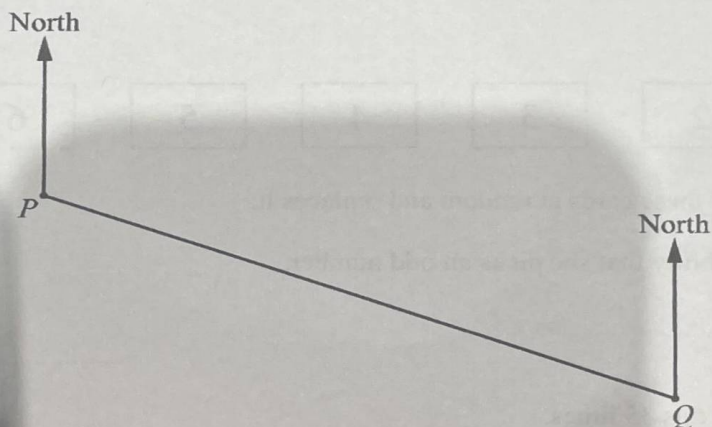


Shade **one** more small square so that the diagram has rotational symmetry of order 2.

[1]

2 The scale drawing shows the positions of two villages, P and Q .

The scale is 1 cm represents 0.5 km.



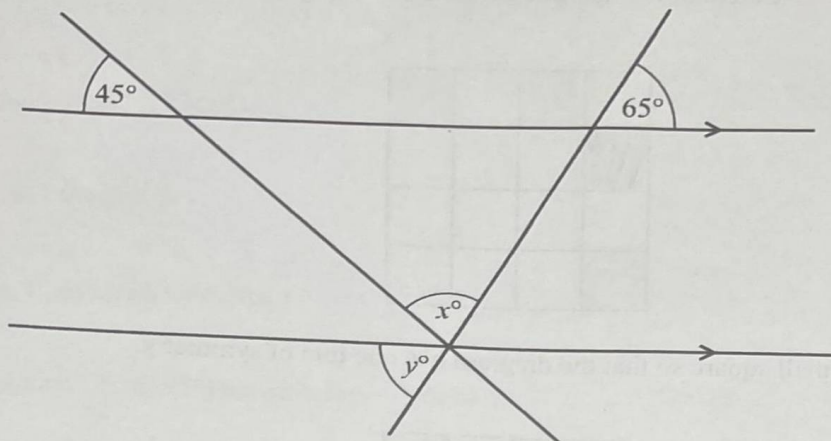
(a) Find the actual distance between village P and village Q .

..... 4.5 km [2]

(b) Measure the bearing of village Q from village P .

..... 109 [1]





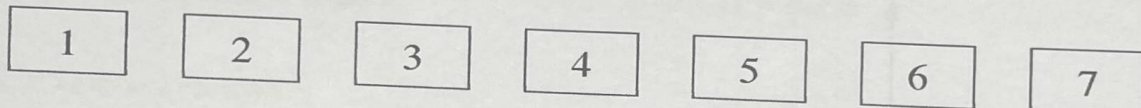
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The diagram shows two straight lines intersecting two parallel lines.
Find the value of x and the value of y .

$x = 70$

$y = 65$

[3]



Samira picks one of these cards at random and replaces it.

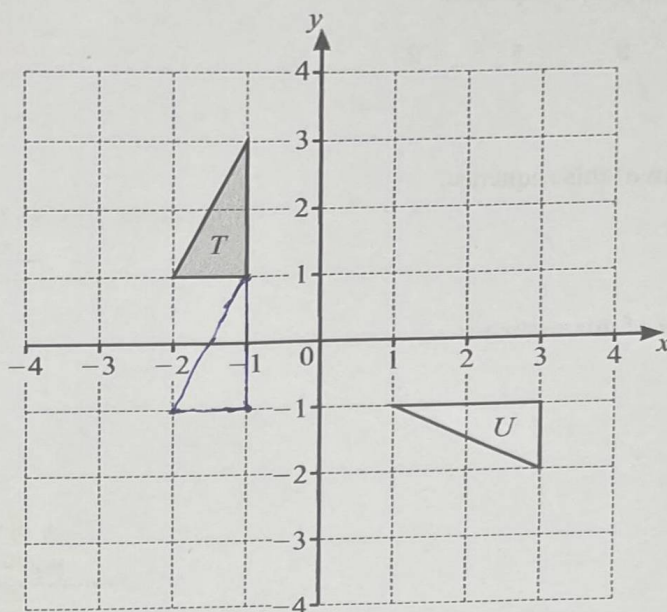
(a) Find the probability that she picks an odd number.

$\frac{4}{7}$ [1]

(b) Samira repeats this 35 times.

Calculate the number of times Samira is expected to pick an odd number.

20 [1]



- (a) Translate triangle T by the vector $\begin{pmatrix} 0 \\ -2 \end{pmatrix}$. [1]

- (b) Describe fully the **single** transformation that maps triangle T onto triangle U .

Rotation 90° anticlockwise at centre $(2, 2)$.

[3]

6 Solve.

(a) $8x + 7 = 39$

$$8x = 32$$

$$x = \frac{32}{8}$$

$$= 4$$

$$x = 4$$
 [2]

(b) $2(5y - 1) = 24$

$$y = 2.6$$
 [3]

- 7 These are the first 4 terms of a sequence.

11 8 5 2

- (a) Find the next term of this sequence.

..... - 1 [1]

- (b) Find the n th term of this sequence.

..... $14 - 3n$ [2]

- 8 Find the highest common factor (HCF) of 36 and 54.

..... 18 [2]

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- 9 A is the point $(3, -1)$.

$$\overrightarrow{AB} = \begin{pmatrix} 2 \\ -4 \end{pmatrix}$$

(a) $\overrightarrow{AC} = 2\overrightarrow{AB}$

Find the coordinates of the point C .

(7 , -9) [2]

- (b) The length of AB is $k\sqrt{5}$.

Find the value of k .

$k =$ 2 [2]

- (c) P is a point on AB .

$$AP : PB = 1 : 3$$

Find the position vector of P .

$\begin{pmatrix} 3.5 \\ -2 \end{pmatrix}$ [2]

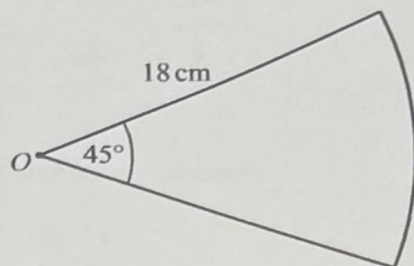
[Turn over]

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SCALE

The diagram shows a sector of a circle, centre O .
The length of the arc is $n\pi$ cm.

Find the value of n .

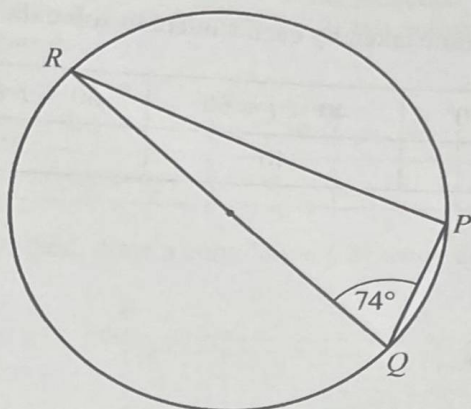
$$n = \dots\dots\dots 4.5 \dots\dots\dots [2]$$

- 11 (a) Write 0.007 08 in standard form.

$$\dots\dots\dots 7.08 \times 10^{-3} \dots\dots\dots [1]$$

- (b) Work out $(3.8 \times 10^{22}) + (3.8 \times 10^{23})$.
Give your answer in standard form.

$$\dots\dots\dots 4.18 \times 10^{23} \dots\dots\dots [2]$$



NOT TO
SCALE

P , Q and R lie on a circle.
 QR is a diameter.

Find angle PRQ .
Give geometrical reasons for your answer.

Angle $PRQ = 16^\circ$ because angle in a semicircle is equal to 90° [2]

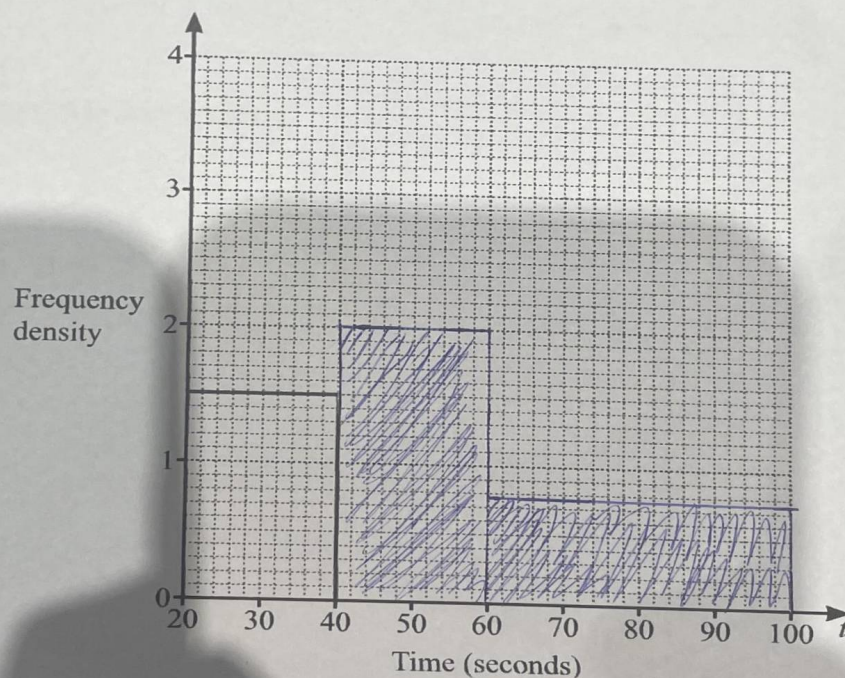
- 13 (a) 100 students solve a puzzle.
The table shows information about the time taken by each student to solve the puzzle.

Time (t seconds)	$20 < t \leq 40$	$40 < t \leq 60$	$60 < t \leq 100$
Frequency	30	40	30

- (i) Work out an estimate of the mean.

..... 53 s [4]

- (ii) Complete the histogram to show the information in the table.

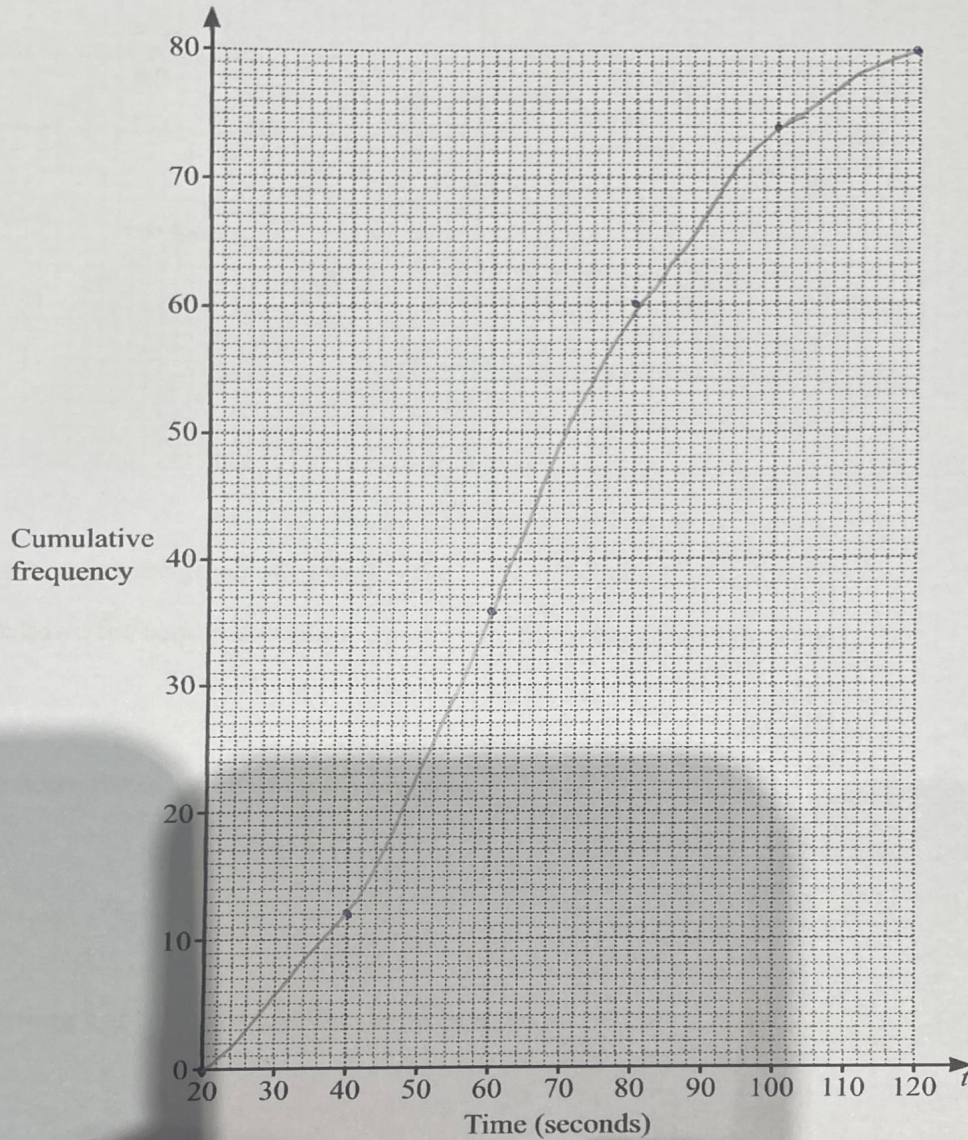


[2]

- (b) 80 adults solve the same puzzle as the students.
The cumulative frequency table shows information about the time taken by each adult to solve the puzzle.

Time (t seconds)	$t \leq 20$	$t \leq 40$	$t \leq 60$	$t \leq 80$	$t \leq 100$	$t \leq 120$
Cumulative frequency	0	12	36	60	74	80

- (i) On the grid, draw a cumulative frequency diagram.



[3]

- (ii) Use your cumulative frequency diagram to find an estimate for

(a) the median

..... 63 s [1]

(b) the lower quartile.

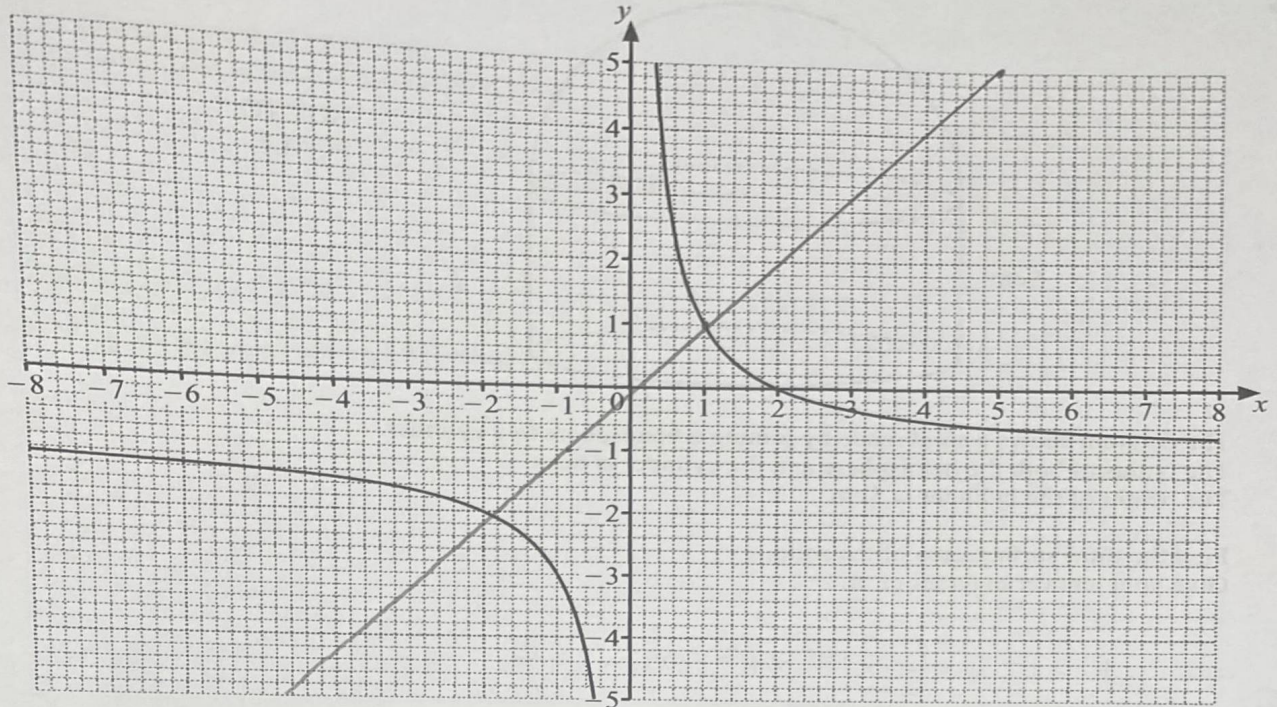
..... 49 s [1]

14 Write 0.25 as a fraction.

$$\frac{23}{90} \quad [2]$$



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The diagram shows the graph of $y = \frac{2}{x} - 1$.

- (a) Write down the coordinates of the point where the graph crosses the x -axis.

(2 , 0) [1]

- (b) Write down the equation of each asymptote.

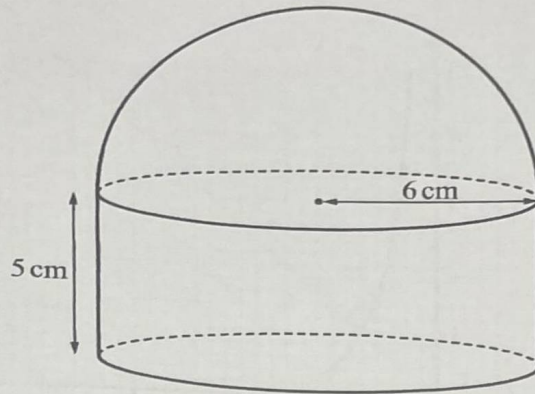
$x = 0$

$y = -1$

[2]

- (c) By drawing a suitable straight line on the grid, solve $\frac{2}{x} - x - 1 = 0$.

$x =$ -2 $\text{ or } x =$ 1 [3]



NOT TO
SCALE

The diagram shows a solid made by joining a hemisphere to a cylinder.
The radius of both the hemisphere and the cylinder is 6 cm.
The height of the cylinder is 5 cm.

Find the **total** surface area of the solid.
Give your answer in terms of π .

$$\pi \times 6^2 + \frac{1}{2} \times 4\pi \times 6^2 + 2\pi \times 6 \times 5$$

$$= 168\pi$$

..... 168π cm^2 [4]

17 Find the value of

(a) $125^{\frac{2}{3}}$

..... 25 [2]

(b) $4^{-\frac{5}{2}}$

..... $\frac{1}{32}$ [2]

18 (a) $\frac{9}{\sqrt{3}}$

Rationalise the denominator.
Give your answer in its simplest form.

$3\sqrt{3}$ [2]

(b) $(5 - \sqrt{2})(1 + 3\sqrt{2}) = c + k\sqrt{2}$

Find the value of c and the value of k .

$c = -1$

$k = 14$ [2]

19 Write as a single fraction in its simplest form.

(a) $\frac{5a}{6} \times \frac{3b}{a}$

$\frac{5b}{2}$ [2]

(b) $\frac{p}{2} + \frac{3t}{4}$

$\frac{2p + 3t}{4}$ [2]

(c) $\frac{2}{x-2} - \frac{3}{x+1}$

$\frac{-x + 8}{x^2 - x - 2}$ [3]

Turn over

20

$$y \propto \frac{1}{\sqrt{x}}$$

- (a) When $x = 9$, $y = 2$.

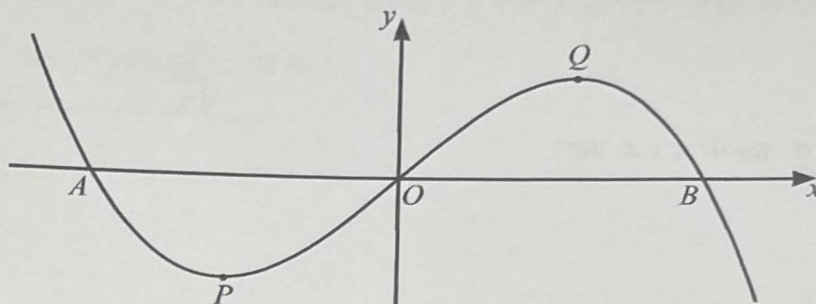
Find the value of y when $x = 36$.

$$y = \frac{1}{2} \dots\dots\dots [3]$$

- (b) When x is increased by a factor of 4, the value of y changes by a factor of p .

Find the value of p .

$$p = \frac{1}{2} \dots\dots\dots [1]$$

NOT TO
SCALE

The diagram shows the graph of $y = 3x - x^3$.
The graph crosses the x -axis at A , at O and at B .
The turning points of the graph are at P and at Q .

- (a) Find the x -coordinate of A and the x -coordinate of B .
Give your answers as exact values.

x -coordinate of A $-\sqrt{3}$

x -coordinate of B $\sqrt{3}$

[3]

- (b) (i) Differentiate $3x - x^3$.

..... $3 - 3x^2$ [2]

- (ii) Find the coordinates of P and Q .

P (..... -1 , -2 )

Q (..... 1 , 2 )

[4]

22 (a) Write down the exact value of $\tan 60^\circ$.

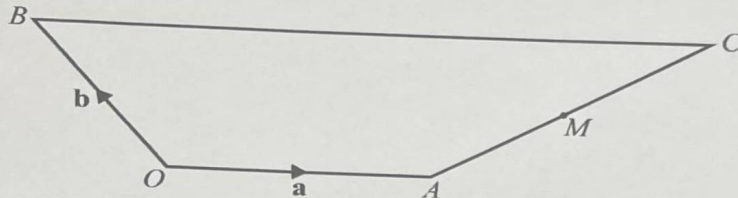
$\sqrt{3}$

[1]

(b) Solve $2 \sin x - 1 = 0$ for $0^\circ \leq x \leq 360^\circ$.

$x = 30$ or $x = 150$ [3]

23



NOT TO
SCALE

In the diagram, OA is parallel to BC .

$BC = 3OA$

M is the midpoint of AC .

The position vector of A is $\mathbf{a}$ and the position vector of B is $\mathbf{b}$.

Find the position vector of M .

Give your answer in terms of $\mathbf{a}$ and $\mathbf{b}$, in its simplest form.

$2\mathbf{a} + \frac{1}{2}\mathbf{b}$

[3]

- 24 The line $y = 7x + 3$ intersects the curve $y = x^2 + 5x - 12$ at the points A and B .

Find the coordinates of A and B .

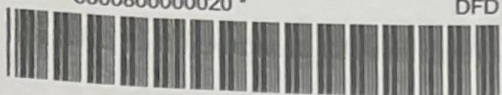
$$A (\dots 5 \dots , \dots 38 \dots)$$

$$B (\dots -3 \dots , \dots -18 \dots)$$

[5]

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